Adil Rahmat Hidayah

Civil and structural engineer

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PROFILE

Civil/Structure Engineer in Jakarta working on oil and gas facilities: foundations for process units, vessels, tanks and pump skids, and structural analysis of shelters, platforms and pipe racks. Graduate of Institut Teknologi Bandung (2025). Builds calculation tools and AI-assisted workflows that take a model from design basis to report and drawings, and reviews every result before it is issued.

EXPERIENCE

Civil/Structure Engineer Sep 2026 – present

PT Tiga Inti Pilar, Jakarta Oil and gas facilities

- Design of a loading pump and truck shelter.
- Design of a closed drain pit and pipe supports.
- Foundation design for a gas scrubber and a vent stack.

Civil and Structure Engineer Jul – Aug 2026

PT Trigan Multi Terapan, Jakarta Early production facilities, AKM Field Development

- Foundation design for process units, separators and storage tanks in AFES under ACI 318-19 and ASCE 7-22.
- Structural analysis of equipment shelters and steel platforms in STAAD.Pro, with connection and anchor bolt design in IDEA StatiCa.
- Foundation and stability design, and lifting analysis, for container-based buildings on precast plinth beams.

Civil Engineer Oct 2025 – Jun 2026

PT Singgar Mulia, Jakarta Midstream infrastructure, Genting FLNG Project

- Pile and shallow foundations for vertical and horizontal vessels, storage tanks and pump skids.
- Pipe racks, equipment shelters and concrete pits in STAAD.Pro, with connections and anchor bolts in IDEA StatiCa.
- Secondary structures including bund walls and pig protection walls.

Engineer Intern Jun – Aug 2024

Patimban Port Development Project, Subang 28.2-hectare container terminal, IDR 5.5T project

- Validated reclamation and berth structures, and used SAP2000 to find and resolve three design conflicts before construction.
- Tracked seven project milestones with Microsoft Project.

SELECTED PROJECTS

FAIR Foundation Design. Web calculator for equipment foundations. Checks stability, footing, pedestal, piles, pile caps and anchor bolts from one set of inputs or from STAAD reactions.

Structural calc to drawing agent. AI agent and scripts that carry a STAAD model through the calculation report, drawings and connection details, with independent checkers for drawings and documents.

Steel shelter on very soft clay. Steel shelter over loading pumps and truck bays. Shallow footings failed, so the foundation became piles with tie beams. Governing ratios 0.95 (rafter, pile compression).

Three vessels on one pile cap. Pile cap for three filter coalescers and a steel platform, checked for pile group, settlement, flexure, shear and crack control.

EDUCATION

Bachelor of Civil Engineering, Institut Teknologi Bandung 2021 – 2025 · GPA 3.58/4.00

Thesis on fixed offshore structure design, covering in-place, seismic, lifting and on-bottom stability analysis of a three-legged jacket in Bentley SACS.

AWARDS AND RECOGNITION

- Grand Champion, Kompetisi Jembatan Indonesia 2024. First for ITB in 19 years, and the team's steel truss bridge was also the lightest (42 kg) and fastest to assemble (54 minutes).
- 1st place, National Tender Competition, CIFEST 2023 · 1st place, Transportation Analysis Competition, 2023
- Ganesha Karsa Awardee, ITB 2023, and TELADAN Scholarship, Tanoto Foundation 2022
- Student exchange at United Arab Emirates University, Future+ Program 2025

SKILLS

Software: STAAD.Pro, Bentley SACS, AFES, IDEA StatiCa, SAP2000, ETABS, SpColumn

Codes: ACI 318-19, ANSI/AISC 360-22, ASCE 7-22, API RP 2A, API RP 2EQ, SNI

Structures: Mat, pile and tank foundations, precast plinth beams, pipe racks, equipment shelters, steel platforms, lifting analysis, offshore jackets

Automation and AI: AI agents and AI-assisted automation for STAAD model, report and drawing generation, Python scripting, Excel calculation toolkits, web-based calculators